

Lenard Dome

Doctoral Candidate
School of Psychology
Plymouth University, UK

✉ lenard.dome@plymouth.ac.uk
🌐 <https://lenarddome.github.io/>
🐱 [lenarddome](#)
🐦 [lenarddome](#)

RESEARCH AREAS

Computational Modelling of the Complexity of Cognition and Behaviour
Categorization, Learning, Social Categorization
Open Science and Open Software

ACADEMIC POSITIONS

2020 - 2023 Ph.D. Psychology. Plymouth University. (Supervisor: Andy Wills; Viva: Sept 2023).
2022 - 2023 Visiting Lecturer, Tutor and Specialist. University of Plymouth.
2019 - 2023 Teaching Assistant. University of Plymouth.
2019 - 2020 M.Sc. Advanced Psychology. Plymouth University. (Supervisor: Patric Bach).
2017 - 2019 Research Assistant. (Line Manager: Andy Wills).
2017 - 2018 Research Assistant. (Line Manager: Marina Wimmer).

QUALIFICATIONS

2019 - 2020 M.Sc. Advanced Psychology. Plymouth University. *Distinction*.
2017 - 2019 B.Sc. Psychology. Plymouth University. *First*.

GRANTS AND AWARDS

2020 **Awarded** [Undergraduate Project Prize](#). Experimental Psychological Society / British Science Association.
2019 **Awarded** Doctoral Training Studentship (1+3). ESRC & SWDTP. £87,510.62.

PUBLICATIONS

preprint **Dome, L.**, Inkster, A.B., Wills, A.J. “Better generalization through distractions.” doi.org/10.31234/osf.io/eskr9.
Dome, L. & Wills, A.J. “g-distance: On the comparison of model and human heterogeneity.”
doi.org/10.31234/osf.io/ygmcj.
accepted **Dome, L.** & Wills, A.J. “Errorless irrationality: removing error-driven components from the inverse base-rate effect paradigm.” *Proceedings of the 45th Annual Conference of the Cognitive Science Society*. Austin, TX: Cognitive Science Society. doi.org/10.31234/osf.io/936bj.

- 2021 **Dome, L.**, Edmunds, C. E. R., Wills, A. J. “SUSTAIN captures category learning, recognition, and hippocampal activation in a unidimensional vs information-integration task.” *Proceedings of the 43rd Annual Conference of the Cognitive Science Society*. Austin, TX: Cognitive Science Society. <https://escholarship.org/uc/item/5r98q3dr>
- Spicer, S. G., Wills, A. J., Jones, P. M., Mitchell, C., & **Dome, L.** “Representing uncertainty in the Rescorla-Wagner model: blocking, the redundancy effect, and outcome base rate.” *Open Journal of Experimental Psychology and Neuroscience*. doi.org/10.46221/ojepn.2021.6623
- 2019 Wimmer, M. C., **Dome, L.**, Wennekers, T. & Hancock, P. J. B. “Is the letter cancellation task a suitable index of ego-depletion? Empirical and conceptual issues.” *Social Psychology*. doi.org/10.1027/1864-9335/a000393

RESEARCH SOFTWARE

-  **psp** *maintainer, developer* Implements an n -dimensional parameter space partitioning algorithm for evaluating the global behaviour of formal computational models as described by Pitt, Kim, Navarro and Myung (2006). 5.7K downloads. [\[source code\]](#) [\[cran\]](#)
-  **catlearn** *senior developer* Catlearn is an archive of formal models of categorization and learning, plus benchmark datasets to test them against. It currently houses 14 formal mathematical models of learning and categorization. I have contributed 8 models to the package. 38K downloads. [\[source code\]](#) [\[cran\]](#)

INVITED TALKS

- 2021 Clearing confounds from the inverse base-rate effect: Irrationality and concurrent load. “2020 British Science Association / Experimental Psychological Society Undergraduate Project Prize Talk.” 2021. [\[video\]](#).
- The role of base rates in categorizing ambiguity and forming stereotypes. “Liverpool John Moores University: Cognitive and Affective Neuroscience Group Seminar.” 2021.

CONFERENCE PRESENTATIONS

- 2023 Experimental Psychology Society Meeting. Plymouth.
- 2022 22nd Associative Learning Symposium: Gregynog Hall, Wales, UK.
- 2021 Cognitive Science Conference: Online.
- 2019 The British Psychological Society: South West Undergraduate Conference. Plymouth, UK.

TECHNICAL SKILLS

Linear and Non-linear Optimisation: *grid-search global optimisation, differential evolutionary algorithms for global optimisation, gradient descent on error.*

Global Model Behaviour, Data Discretization: *n -dimensional Parameter Space Partitioning.*

Computational Modelling, A.I.: *Embedding, Sequence Modelling, Feed-forward and Convolutional Neural Networks, Adaptive and Static Clustering Models.*

Data Science: *Bayesian Parameter Estimation and Model Comparison, Bayesian Linear Modelling, Supervised clustering, String Metrics (e.g. Levenshtein distance), Bayes Factor, Markov Chain Monte Carlo Sampling Methods, Parametric and Non-parametric Frequentist Methods, Big Data.*

Experimental Design and Data collection: *Designing, implementing, testing and deploying large-scale behavioural experiments through Sona, Prolific, Jatos, PsychoPy.*

PROGRAMMING AND COMPUTING

R: *computational modelling, data analysis, programming, algorithms, statistical analysis.*

Python: *experimental design, programming, computational modelling, statistical analysis.*

C++: *computational modelling, statistical analysis, Rcpp and RcppArmadillo*

Javascript: *jsPsych for experimental design.*

CSS: *webdesign.*

UNIX: *Bash, CLI, Ubuntu Server, ssh.*

High Performance Computing: *Slurm, Microsoft Azure, Google Cloud Computing.*

TEACHING AND PEER-TRAINING EXPERIENCE

How to write clean R code This session was delivered as part of a monthly training session in R that is organised and maintained by the local R community.

Estimating Sample Size with Bayes Factors This workshop is a user-friendly and hands-on introduction to estimating sample sizes with a combination of Bayes Factor and Monte Carlo methods.

Better Tables and Better Graphs A how-to on correlation matrices, custom table of descriptive statistics, and publication-quality graphs showing both central tendency and variability (or uncertainty) of your data.

Introduction to Quantitative Analysis with R This is a half-day session (4 - 5 hours) aimed at first-year Doctoral Clinical Psychology students.

Intermediate Guide to Research Methods in R This is an intermediate half-day session (4 - 5 hours) aimed at second-year Doctoral Clinical Psychology students.

SUPERVISORY EXPERIENCE

2020-2023 **Supervision of Undergraduate Dissertations.**

2019-2023 **Peer-to-peer training and support.**

- Assist, train, and support junior lab members in experimental design, theory, and coding both in R and Python.
- Train and support Ph.D. and M.Sc. students in statistical analysis, experimental design and in the adoption of Open Science methods.